

Matplotlib

Complete Guide

For AI / ML / DL Professionals

30 Concepts

Python

Research Assistant: **Muhammad Huzifa**

Supervisor: **Dr. Muhammad Sajjad**



What is Matplotlib?

Definition

Matplotlib is Python's most popular data visualization library — like a drawing tool for your data that creates charts on screen.

Real-Life Example

A doctor visualizing a patient's heart rate over 24 hours uses a line chart — Matplotlib creates this from raw data.

3 AI/ML Scenarios

Stock prices over time

Temperature changes daily

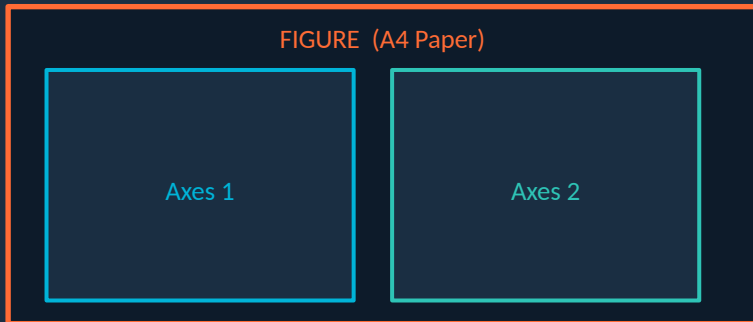
Student test score trends

Exercise

```
import matplotlib.pyplot as plt
plt.plot() # Creates empty plot
plt.show() # Shows the plot
```

Figures & Axes • Line Plot

Figure vs Axes



Line Plot (Time Series)

Best for: Trends over time

Model loss during training

Accuracy over epochs

Stock price predictions vs actual

```
# Concept2:Subplots          # Concept3:Line Plot
fig, axes = plt.subplots(2, 1)  x = [1, 2, 3, 4, 5]
                                y = [1, 4, 9, 16, 25]
plt.show ()                   plt.plot(x, y, color="blue", label="Square")
                                plt.title("Square Numbers"); plt.show ()
```

Scatter Plot • Bar Plot

Scatter Plot

Shows individual data points as dots.
Best for finding patterns & relationships.

AI/ML Uses:

- K-Means clustering results
- Feature correlation checks
- Outlier identification

```
plt.scatter(x, y, c="blue")  
plt.title("Clustering Results")  
plt.show()
```

Bar Plot

Rectangular bars for comparing different groups or categories.

AI/ML Uses:

- Compare ML model accuracy
- Feature importance display
- Class distribution in dataset

```
plt.bar(['Apple', 'Banana', 'Orange'],  
        [50, 30, 45])  
plt.title('Fruit Quantities')
```

Histogram • Pie Chart

Histogram

Groups numbers into bins & shows frequency.
Reveals how data is spread (distribution).

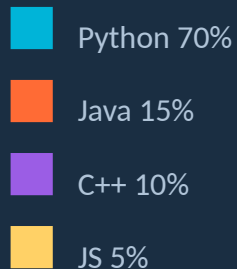
Example:

Exam scores of 100 students.
Pixel intensity in images.
Age distribution in dataset.

```
plt.hist(data, bins=30,  
         edgecolor="black")  
plt.title("Distribution")
```

Pie Chart

Shows parts of a whole as slices.
Best for proportions & percentages.



```
plt.pie(usage, labels=languages,  
        autopct='%1.1f%%')  
plt.title('Language Usage')
```

Colors • Labels • Figure Size

Colors

- ▶ `color="red" / "blue"`
- ▶ HTML hex: `"#FF6B35"`
- ▶ Named: `"tomato", "navy"`
- ▶ `Train=blue, Val=red`

```
plt.plot(x,y,color="red")
plt.plot(x,z,color="blue")
```

Labels

- ▶ `plt.xlabel("Epochs")`
- ▶ `plt.ylabel("Accuracy")`
- ▶ `plt.title("My Model")`
- ▶ `plt.legend()` ← Key!

```
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
```

Figure Size

- ▶ `figsize=(10,6)` inches
- ▶ `dpi=100` → quality
- ▶ `dpi=300` → publication
- ▶ Poster: (20,15)

```
plt.figure(
    figsize=(10,6),
    dpi=100)
```

Concepts 11-15

11

Subplots

Multiple charts in one figure: `fig, axes = plt.subplots(2, 2)`

12

Markers & Lines

Symbols & line styles: `plt.plot(x, y, 'ro--', bs-)`

13

Annotations

Mark important points with arrows & text using `plt.annotate()`

14

Saving Figures

Export to PNG/PDF: `plt.savefig('plot.png', dpi=300)`

15

Grid & Spines

Clean look: `plt.grid(True, '--') + hide top/right spines`

Advanced Visualization Techniques

16 Twin Axes

Two Y-axes on same chart for different scales

```
ax2 = ax1.twinx()
```

17 Heatmaps

Confusion matrix, correlation
attn weights in transformers

```
plt.imshow(data, cmap="hot")
```

18 3D Plots

Decision boundaries, PCA,
loss landscape visualization

```
ax.plot_surface(X, Y, Z)
```

19 Custom Ticks

Control axis labels:
epochs, % formatting, rotation

```
plt.xticks([1, 2, 3],  
            ['Ep1', 'Ep2', 'Ep3'])
```

20 Styles & Themes

Built-in professional themes
for instant beautification

```
plt.style.use(  
    'seaborn-v0_8-darkgrid')
```

Available Styles

- seaborn-v0_8-darkgrid
- ggplot
- fivethirtyeight
- bmh
- dark_background

Hour 3: AI/ML/DL Applications

Concepts 21-25: Real AI/ML Visualizations



21 Training Curves

Plot train vs val loss & accuracy over epochs. Shows overfitting, underfitting, early stopping point.



22 Confusion Matrix

Heatmap of correct vs wrong predictions. Essential for classification model evaluation.



23 Feature Importance

Horizontal bar chart showing which features most affect model predictions. Critical for explainability.



24 Learning Rate Finder

Plot loss vs LR on log scale. Find optimal learning rate — too fast or too slow = bad training.



25 ROC Curve

True Positive Rate vs False Positive Rate. AUC score measures classifier performance for binary tasks.

Training Curves Visualization

Loss Curves

Train Loss —————

ValLoss - - - - -

⚠ Overfitting starts here
(val loss increases, train decreases)

Good model zone: both curves decreasing

Epochs →

Complete Training Visualization

```
fig, (ax1, ax2) = plt.subplots(
    1, 2, figsize=(12, 4))
```

```
ax1.plot(epochs, train_loss,
         "b-", label="Train Loss")
```

```
ax1.plot(epochs, val_loss,
         "r-", label="Val Loss")
```

```
ax1.set_xlabel("Epochs")
```

```
ax1.set_ylabel("Loss")
```

```
ax1.set_title("Model Loss")
```

```
ax1.legend()
```

```
ax1.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```

Concepts 26-30



26 Animations

FuncAnimation for moving charts — show storm movement, training progress, data changes over time.



27 Real-time Plots

plt.ion() for live updating plots. Perfect for live sensor data, stock tickers, heart rate monitors.



28 Decision Boundaries

plt.contourf() + scatter() to show how ML models separate classes in 2D feature space.



29 Error Bars

Show uncertainty with yerr parameter. Confidence intervals for model comparison across multiple runs.



30 ML Dashboard

Complete dashboard: Feature Importance + Confusion Matrix + ROC Curve + Metrics + Learning Curve.

⚡ Quick Reference Card

Basic Plots

```
plt.plot(x,y)
```

```
plt.scatter(x,y)
```

```
plt.bar(x,y)
```

```
plt.hist(data,bins=30)
```

```
plt.pie(sizes)
```

```
plt.barh(y,x)
```

Customization

```
plt.title("Title")
```

```
plt.xlabel("X Label")
```

```
plt.ylabel("Y Label")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.xlim(0,10)
```

Figure & Save

```
fig,ax=plt.subplots()
```

```
plt.subplots(2,2)
```

```
plt.tight_layout()
```

```
plt.savefig('f.png')
```

```
dpi=300 # HQ
```

```
plt.show()
```

Congratulations!

You have mastered Matplotlib for AI/ML/DL



10 Core
Concepts

10 Advanced
Techniques

5 AI/ML
Applications

5 Pro
Skills

Keep practicing — great visualizations make your AI/ML work truly stand out! t